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Conversational Agents (previously Dialogflow CX)

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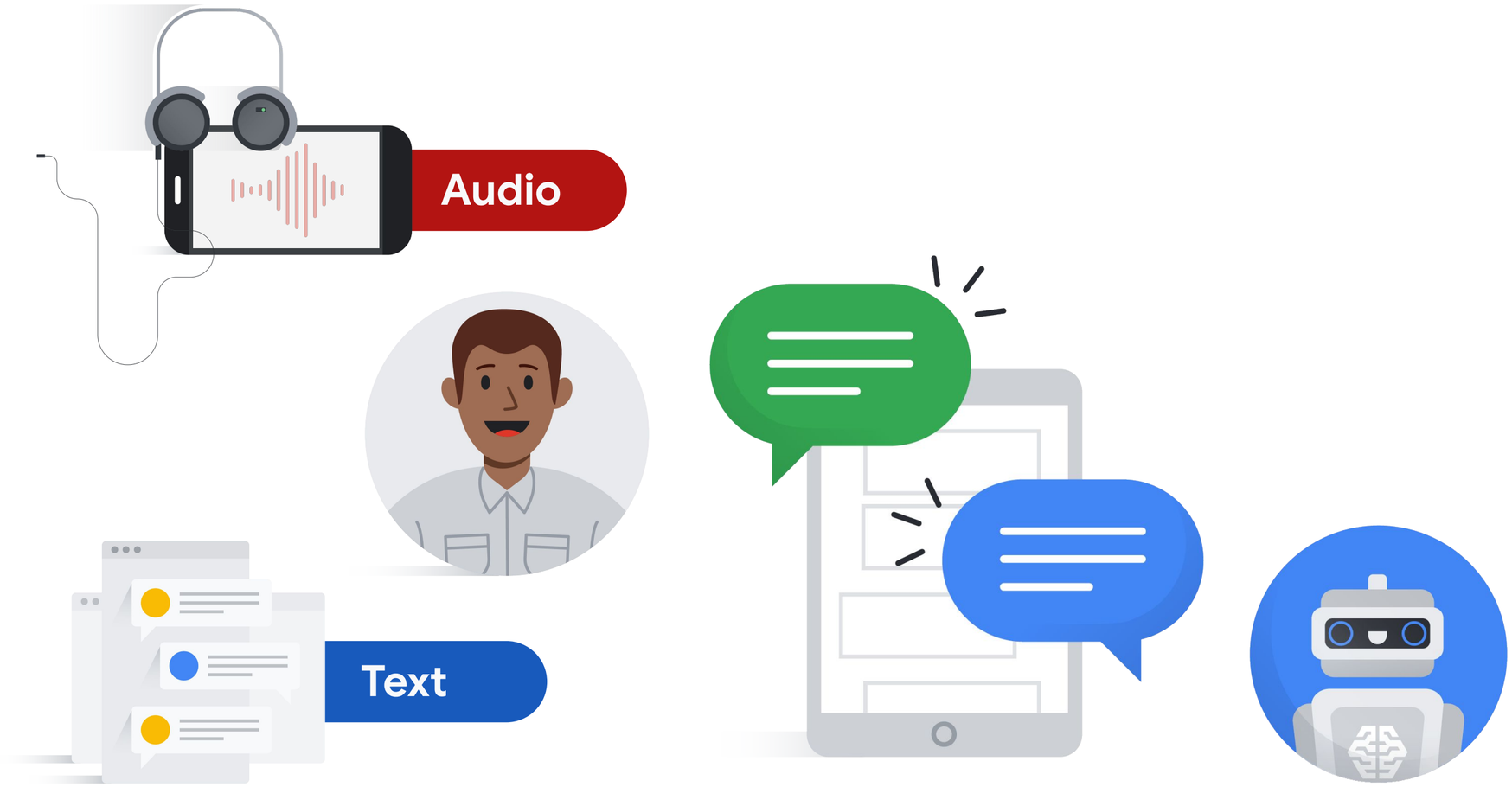
Conversational Agents works with text and voice



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Conversational Agents is now super-charged with Gen AI

Natural conversations

Informational

Responsibly Controlled

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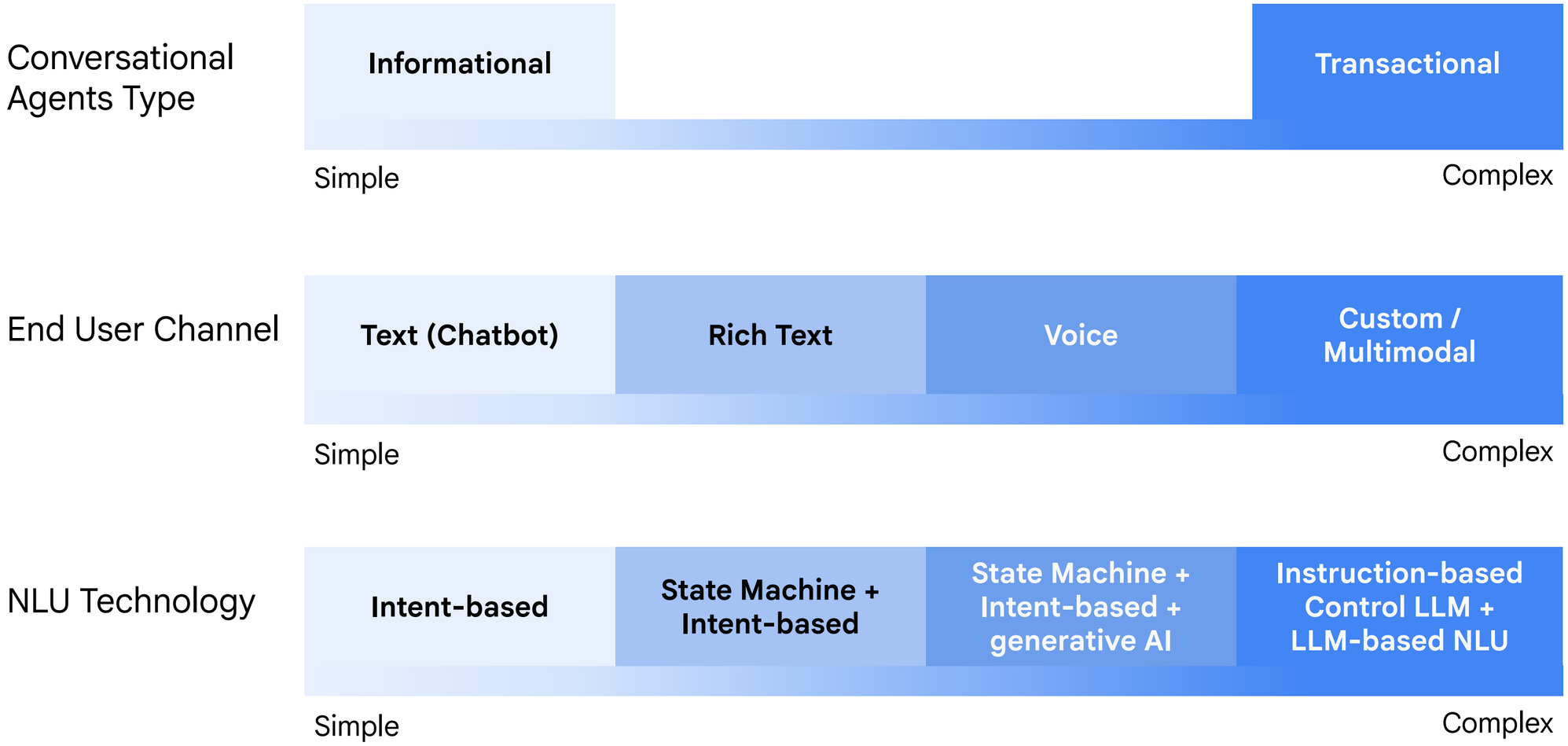
Types of conversational AI solutions



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Types of conversational AI solutions

Conversational Agents Type

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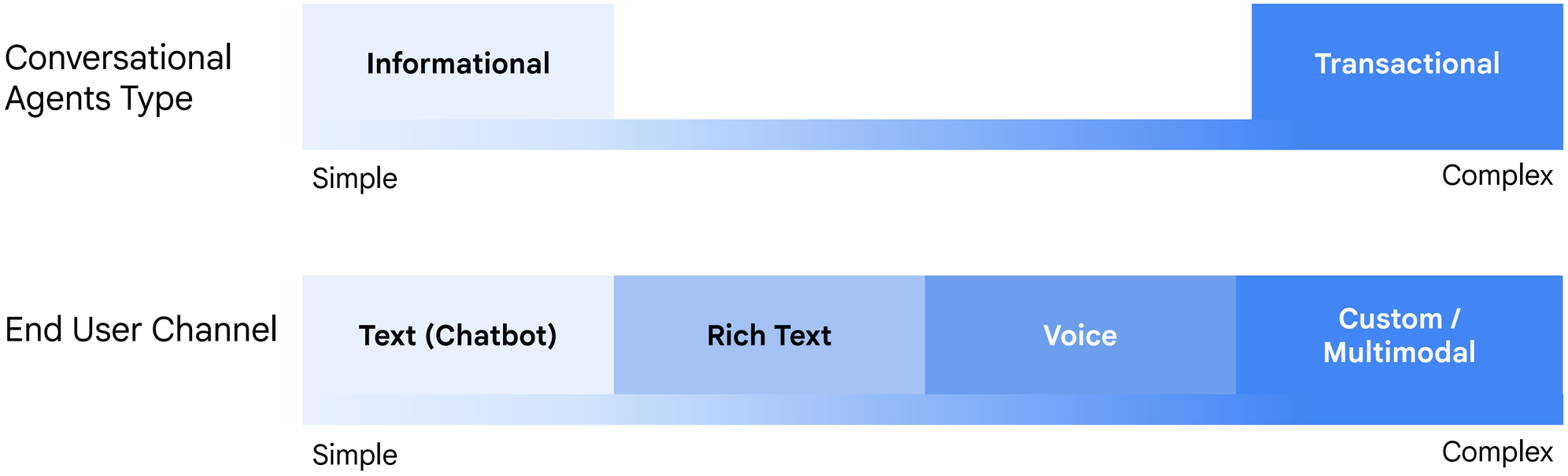
Types of conversational AI solutions



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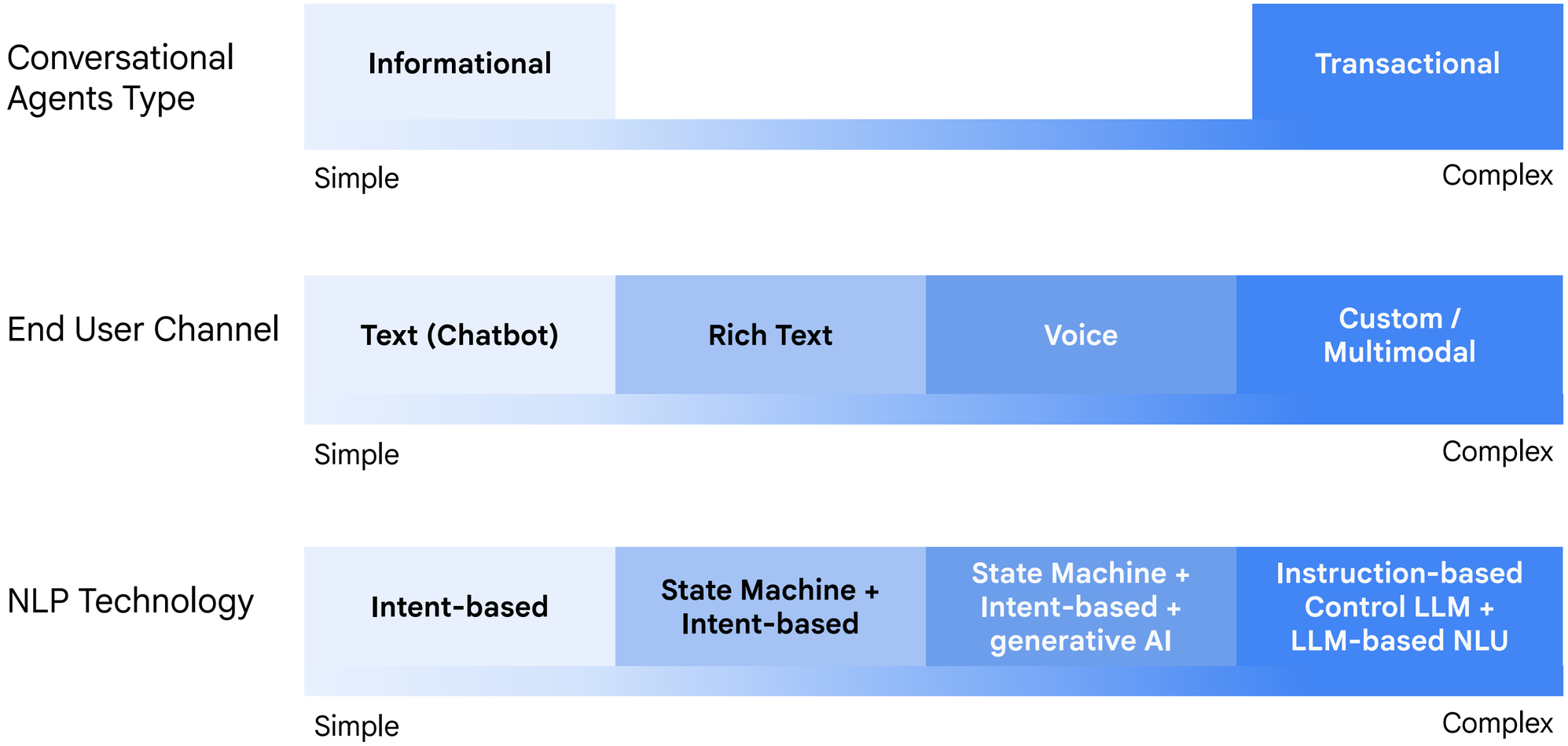
Types of conversational AI solutions



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Generative, deterministic and hybrid conversational agents

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An application that reasons on how to best achieve a goal based on inputs and tools at its disposal.

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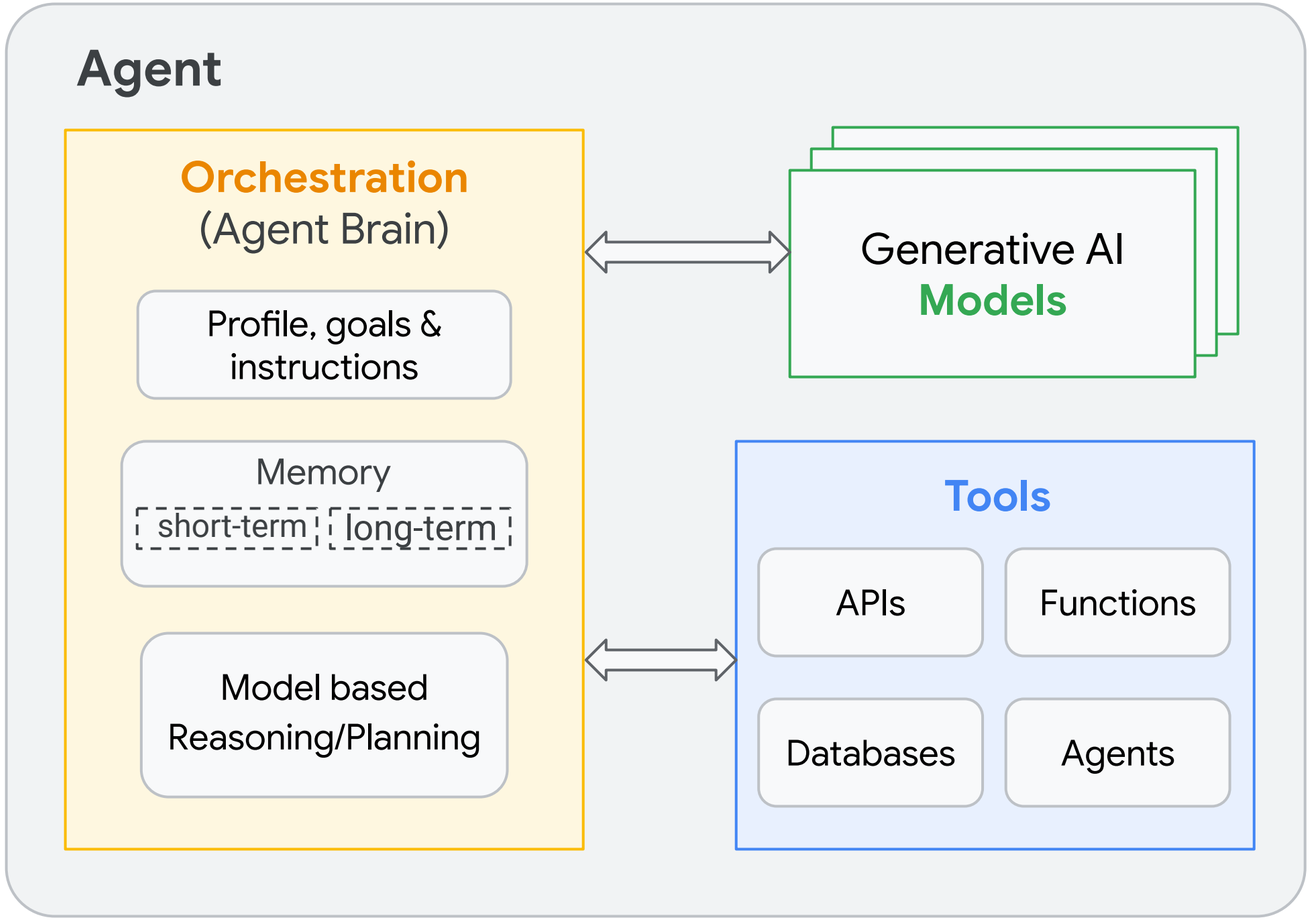
What is an “AI Agent”?



Key Components

- **Orchestration:** Maintain memory and state (including the approach used to plan), tools, data provided and fetched.
- **Models:** Used to reason over goals, determine the plan and generate a response.
- **Tools:** Fetch data, perform actions or transactions by calling other APIs or services.

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Conversational Agents spectrum

Deterministic | Hybrid | Generative

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Conversational Agents spectrum

Deterministic	Hybrid	Generative
		LLM-driven Agents defined with conversational prompting and examples. Reason through future options to assess the impact of these actions on their goals. • RAG-based agents that rely on knowledge bases • FAQ agents • Agents with multiple data connectors • Breadth of use cases OOTB

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Generative agents are built using Playbooks

Playbooks

- Build chat and voice experiences
- Write instructions with natural language
- Playbooks supplement traditional methods

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When to use Playbooks?

- Ease of building and maintaining steering / routing agent
- Handling edge cases
- FAQ and document search where format isn’t critical
- Complex reasoning
- High customization
- Conversation amendments

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Playbooks

DMV Steering | Version history | Save

Basics | Examples | Settings

Playbook name*
DMV Steering

A playbook is the basic building block of a Vertex AI Conversation app. Each playbook is defined to handle specific tasks. Learn more

Goal

Goal*
Your goal is to collect the customer's request and route the customer to the correct service among the variety of online services available from the Department of Motor Vehicles (DMV).

High level description of the goal the playbook intends to accomplish. Learn more

Instructions | Sample

Instructions

```
- Greet the customer, tell them you can help with renewing
driver's licenses and booking appointments at field offices,
ask them how you can help.
- DO NOT attempt to help the user directly.
- ALWAYS transfer them directly to another topic.
- There are 2 topics you can choose from:
    - For driver's license renewals transfer to ${PLAYBOOK:
    Renew Driver's License}
    - For booking appointments at local field offices transfer
    to ${PLAYBOOK: Book Appointment}
- For topics not related to the above topics or sensitive
topics like fees and pricing, transfer to ${PLAYBOOK: Fallback}
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Available tools

This playbook can use selected tools to generate responses. You can also call other tools, including 3P, directly in steps. Create a data store tool to allow this playbook to answer questions using data store content.

+ Data store | Manage all tools

- [ ] dmv_renew_tool (OpenAPI)
- [ ] dmv_book_appointment_tool (OpenAPI)
- [ ] code-interpreter (Extension)

Output parameters [private preview]

DMV Steering | Version history | + Example

Basics | Examples | Settings

Search

Display name	Selection strategy	Language
book only	Auto	
book-appointment-fallback-cancelled	Auto	
fallback-fees	Auto	
fallback-other	Auto	
renew and book	Auto	
renew-fallback-cancelled	Auto	
renew-not-success and book	Auto	
renew-online-success	Auto	

Edit example | Save | Cancel

Display name*
book only

Example info

Each playbook should have one or more examples. These examples are sample conversations between an end-user and the playbook, including the dialogue and actions performed by the playbook. These are effectively few-shot prompt examples for the LLM. Learn more

Selection strategy
(•) Auto ( ) Always ( ) Disabled

Language
Select a language

End example with playbook output information

Summary
Customer booked an appointment.

Summary of the playbook execution result

Playbook state

Playbook output parameters

Name	Value
Add parameter	

- Hi
- Hi, I'm your DMV assistant. I can help you renew your driver's license online and book an appointment at a DMV field office. How can I help you today?
- I want to book an appointment at field office.
- Book Appointment (Playbook)
- Is there anything else I can help with?
- No.
- Thank you! Have a nice day!

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Playbooks

Goal

Instructions

Tools

DMV Steering | Version history | Save

Basics | Examples | Settings

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DMV Steering | Version history | + Example

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fallback-other	Auto	
renew and book	Auto	
renew-fallback-cancelled	Auto	
renew-not-success and book	Auto	
renew-online-success	Auto	

Edit example | Save | Cancel

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Examples

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Playbook Tools

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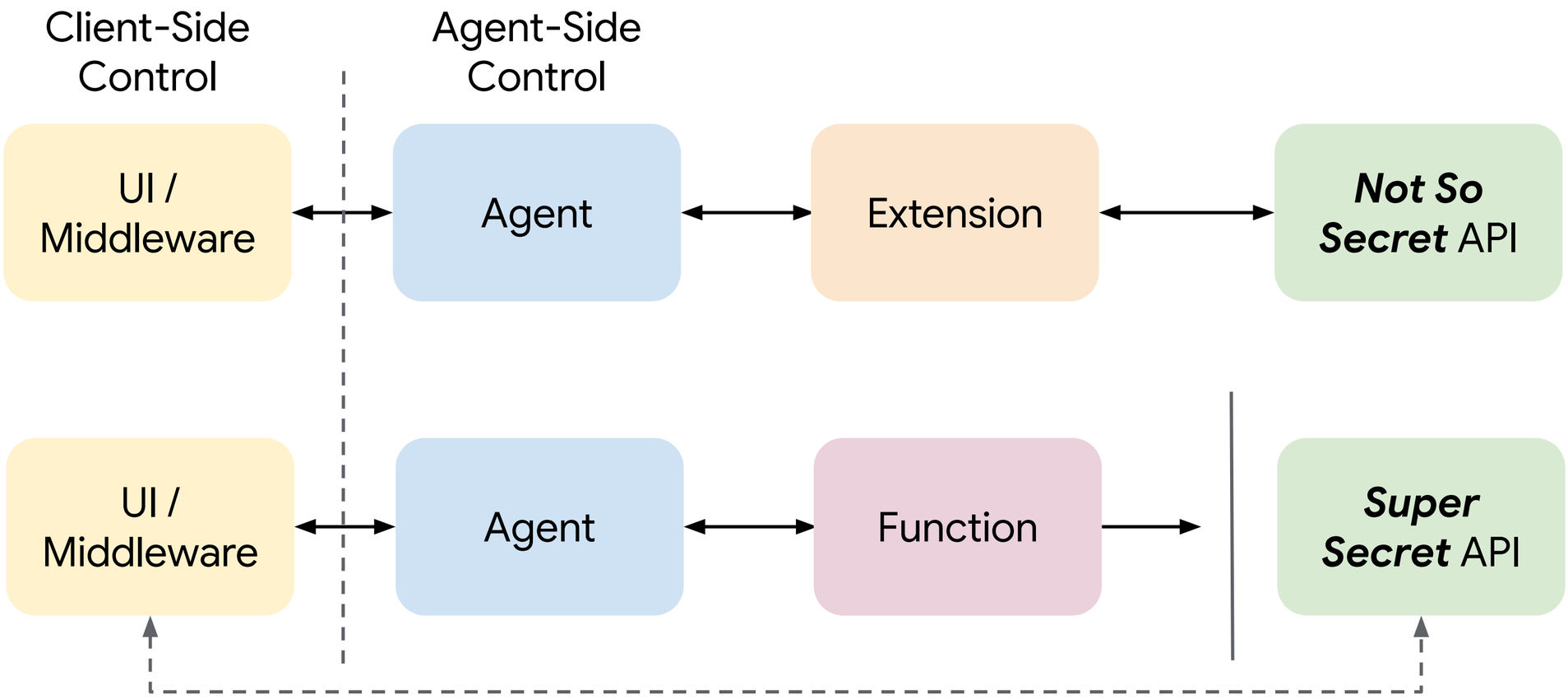
OpenAPI Tools (Extensions) and Function Tools



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Conversational Agents spectrum

Deterministic	Hybrid	Generative
Workflow-based agent that has predefined paths and actions. Typically uses event driven behavior and offers higher control and predictability. • Verbatim responses (legal disclaimers, welcome messages, etc) • Strict ‘if-else’, event- or parameter-based logic • Step-by-step instructions		

Deterministic conversations are built using Flows

Deterministic flows

- Traditional virtual agent construction
- Define flows, pages, intents, and transitions
- Generative AI features can be added

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When to use Deterministic Flows?

- Ensure accuracy in compliance-related use cases
- Implement step-by-step business logic
- Adhere to specific formatting requirements

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Conversational Agents spectrum

Deterministic	Hybrid	Generative
Flows		Playbooks

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Conversational Agents spectrum

Deterministic	Hybrid	Generative
Flows		Playbooks

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Conversational Agents spectrum

Deterministic

Flows

Hybrid

Complex Agents designed to ‘steer’ users between various use cases, provide flexible and adaptive conversational handling.

- Add generative features or switch between deterministic and generative

Generative

Playbooks

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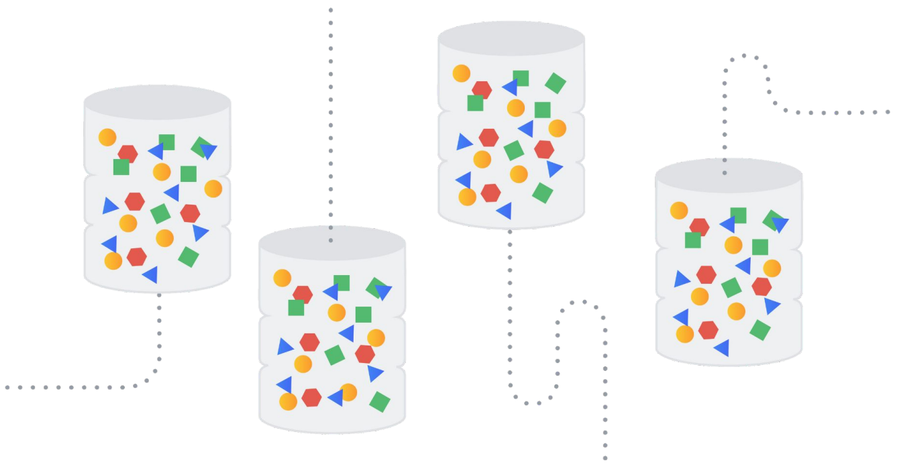
Data stores



- Use data stores from Playbooks and Flows
- Answers grounded in your data
- Enables RAG for external or proprietary data

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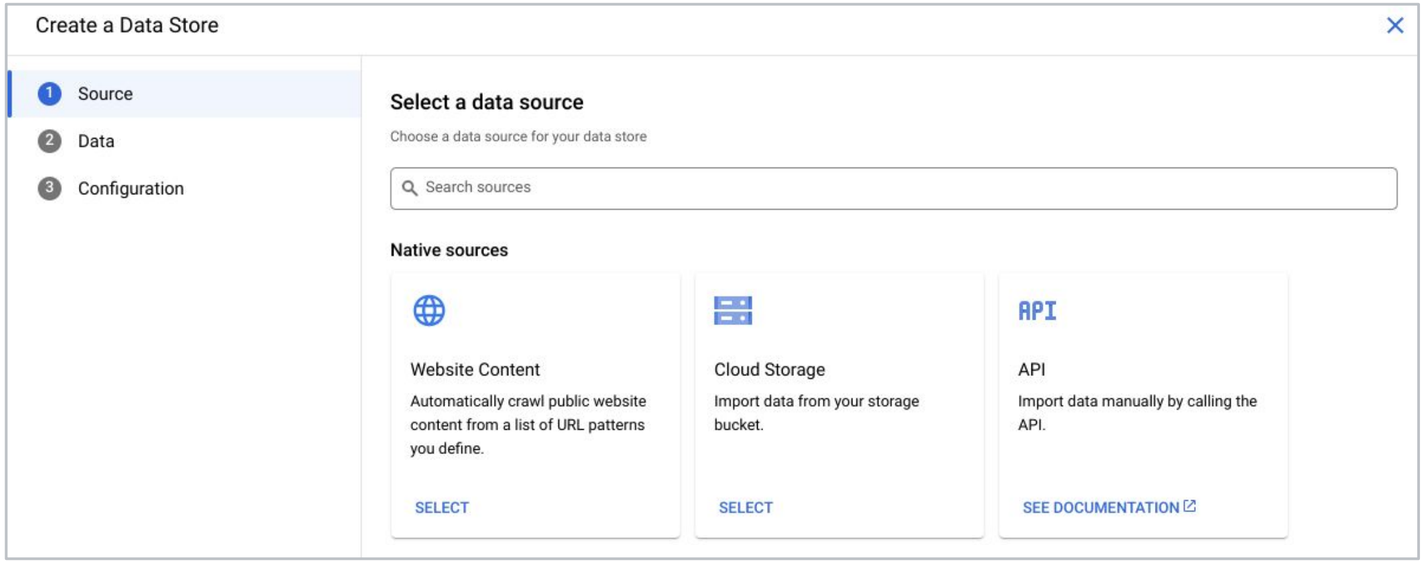
Data stores

Currently supported sources:

- Website URLs
- BigQuery
- Cloud Storage



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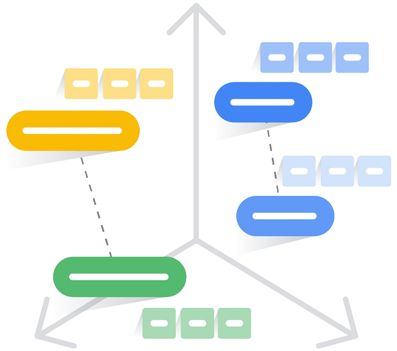
Generative AI features for deterministic Flows



Generators

Generative Fallback

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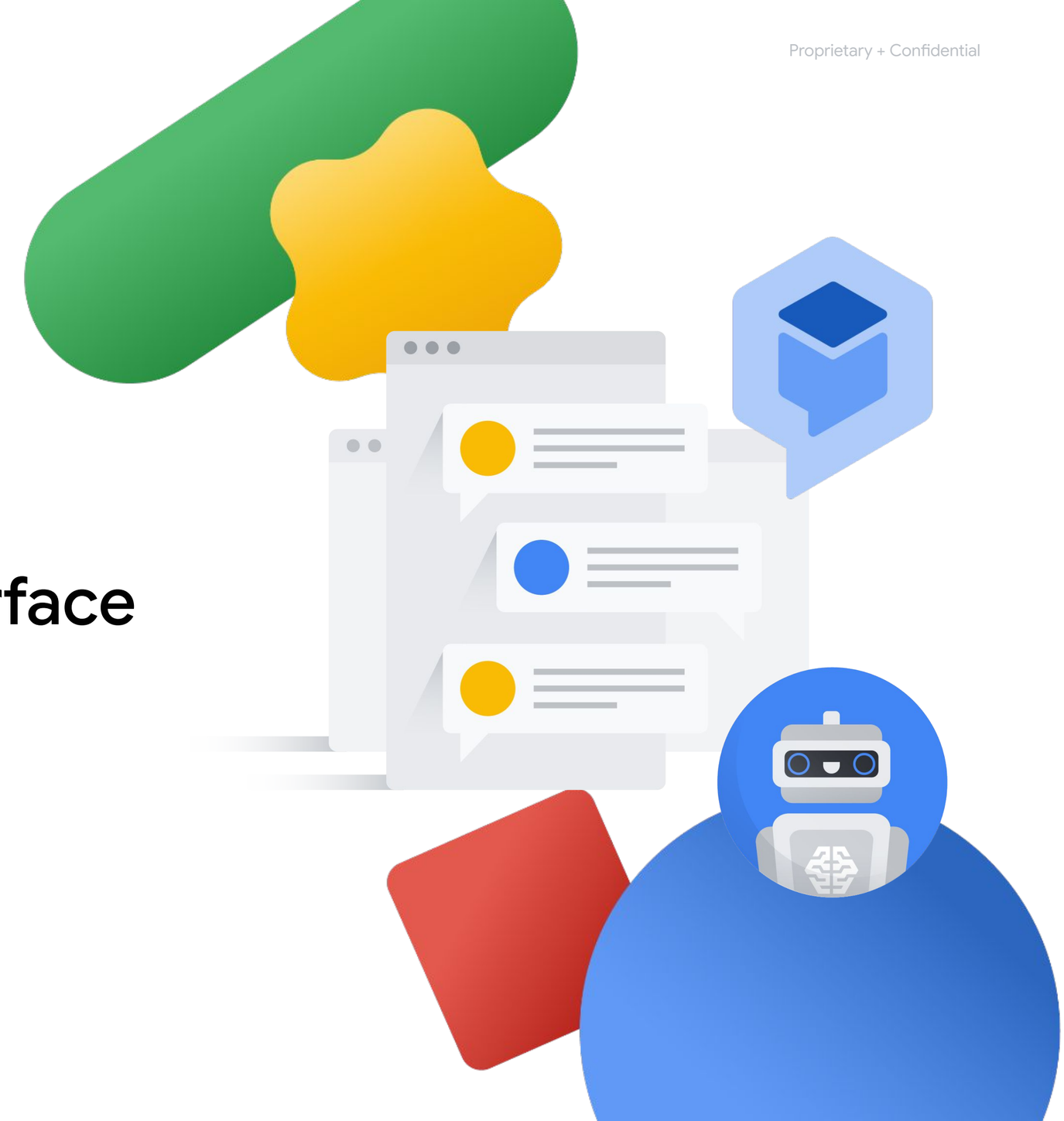
Conversational Agents architecture and user interface



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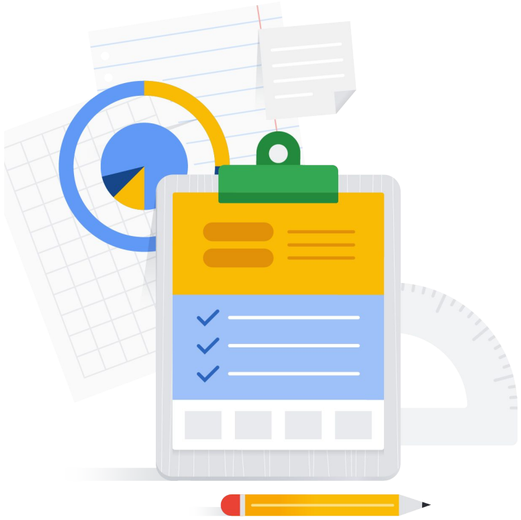
Conversational Agents is Enterprise-ready

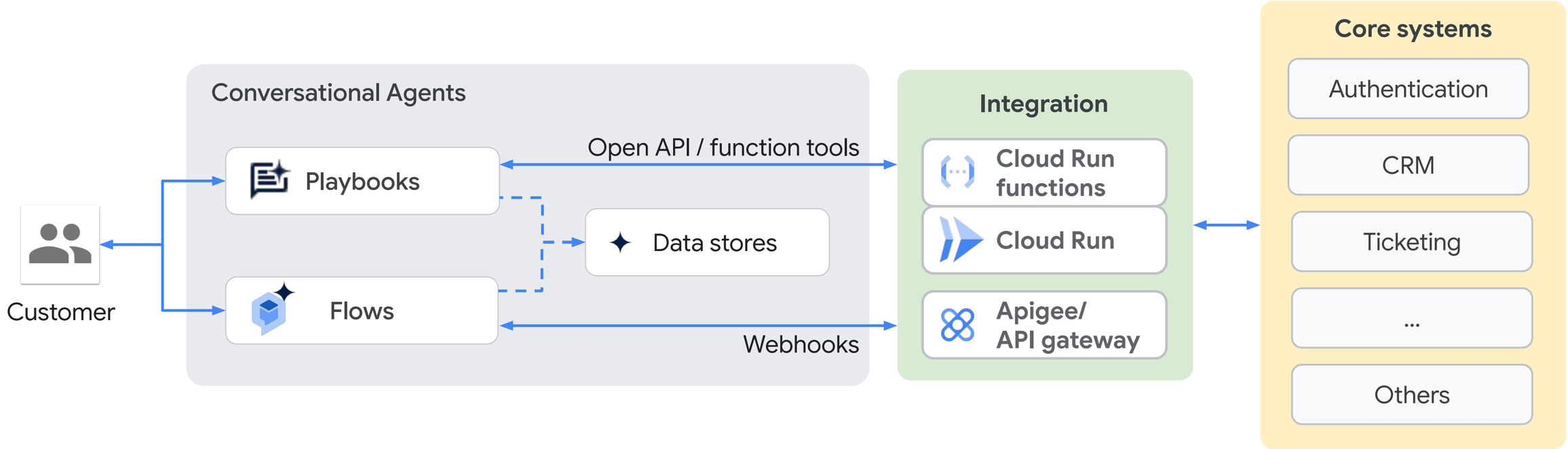
Scalability

Security



Reliability and resilience

High-Level virtual agent architecture with Generative AI



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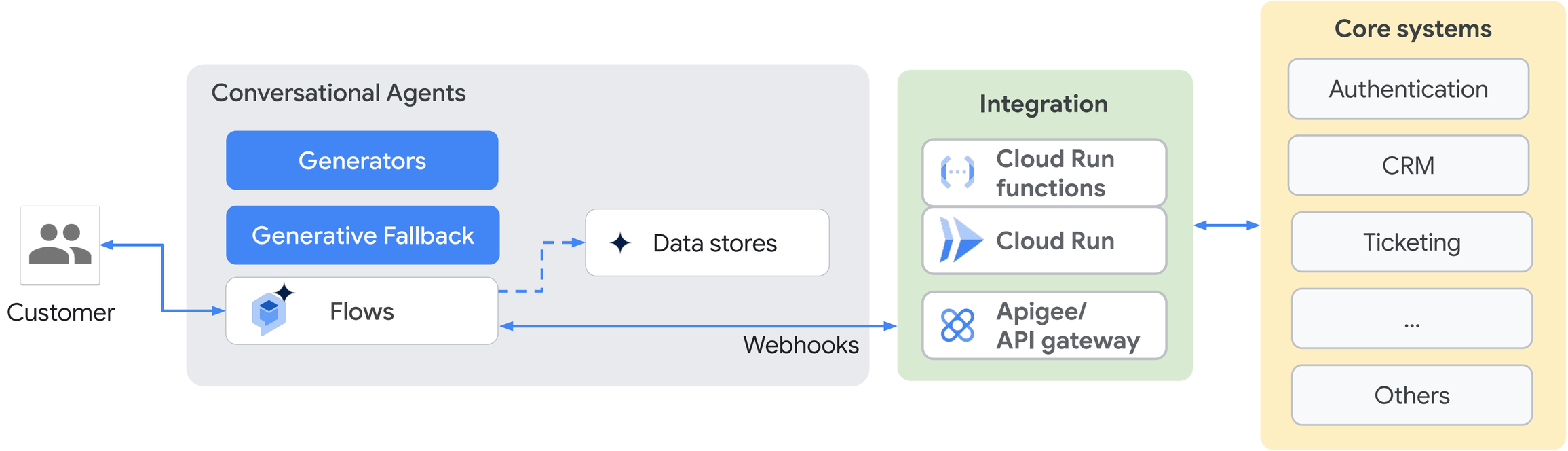

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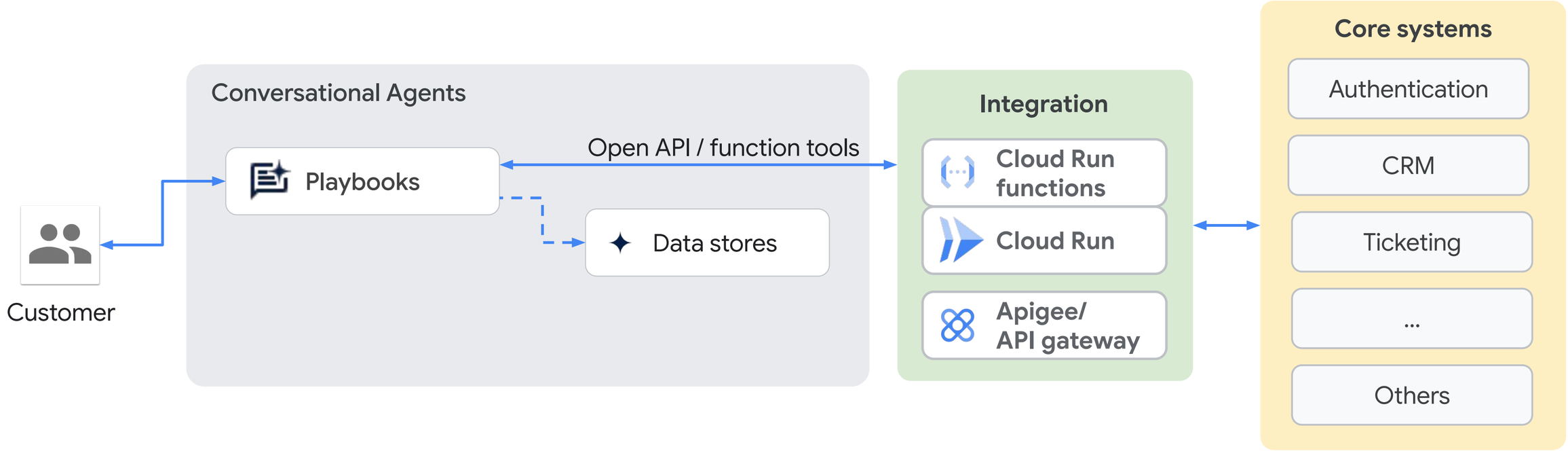

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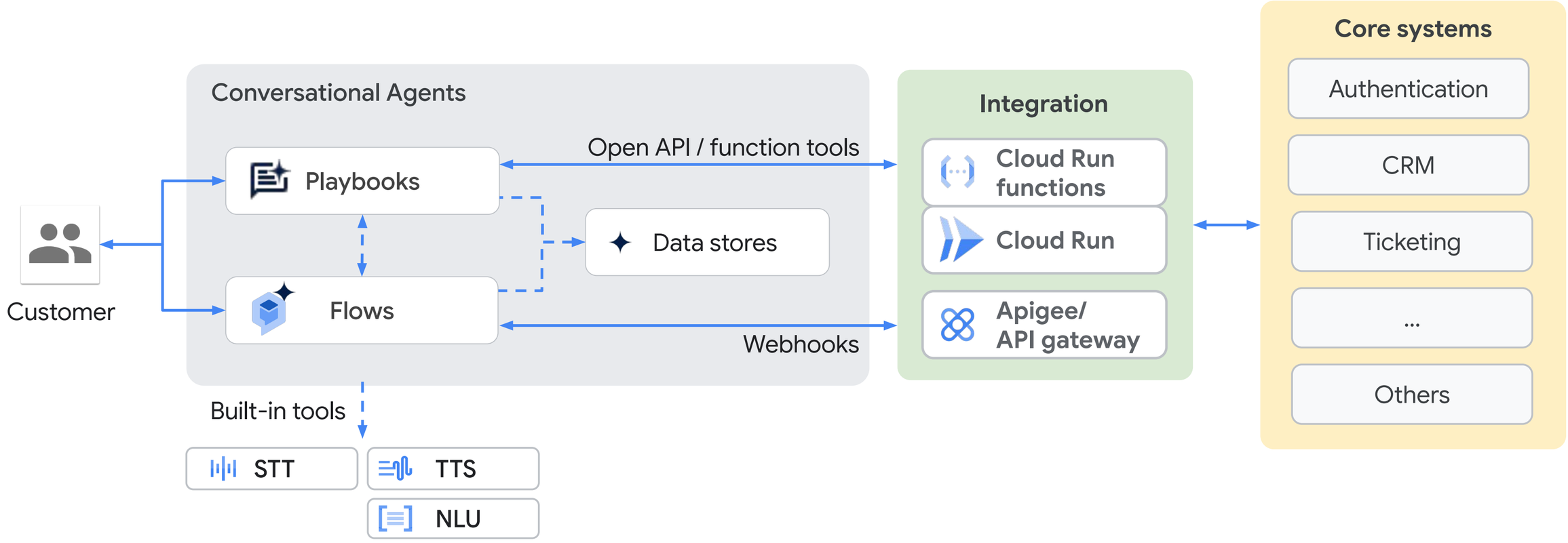

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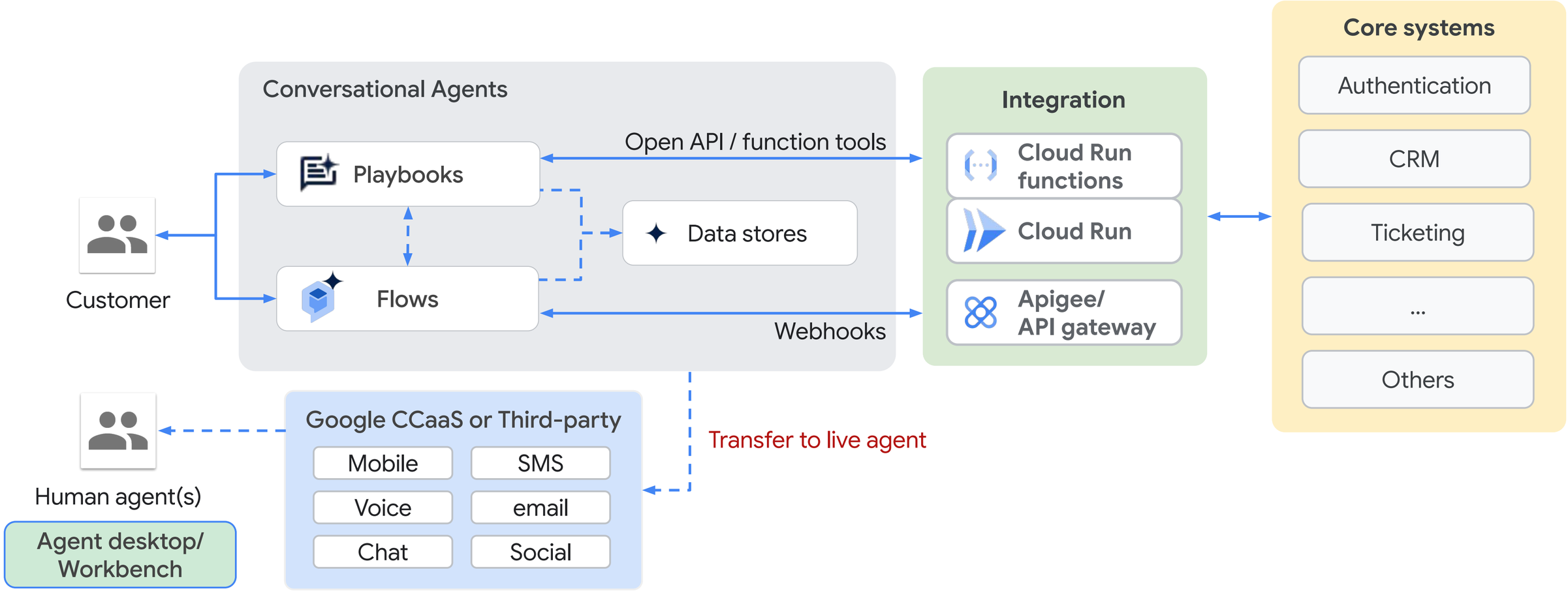
High-Level virtual agent architecture with Generative AI



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How do you create Conversational Agents in the Google Cloud Console?

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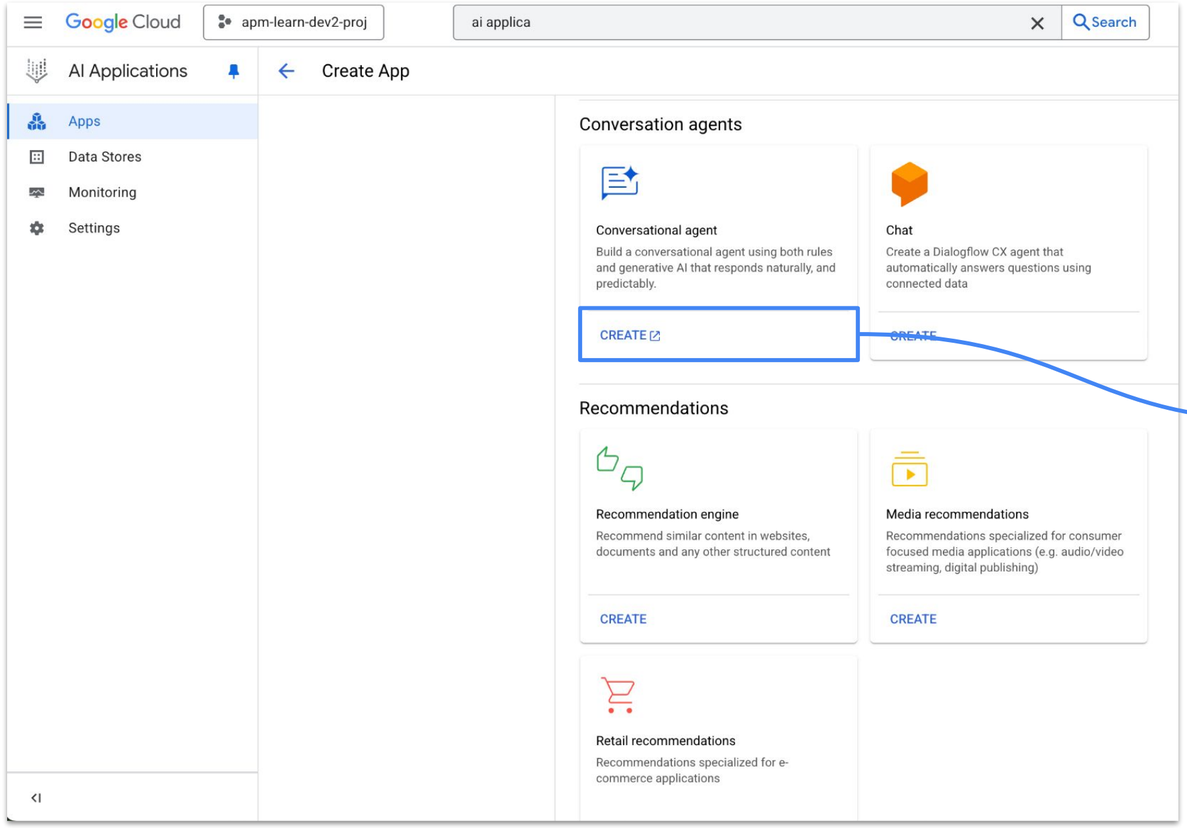
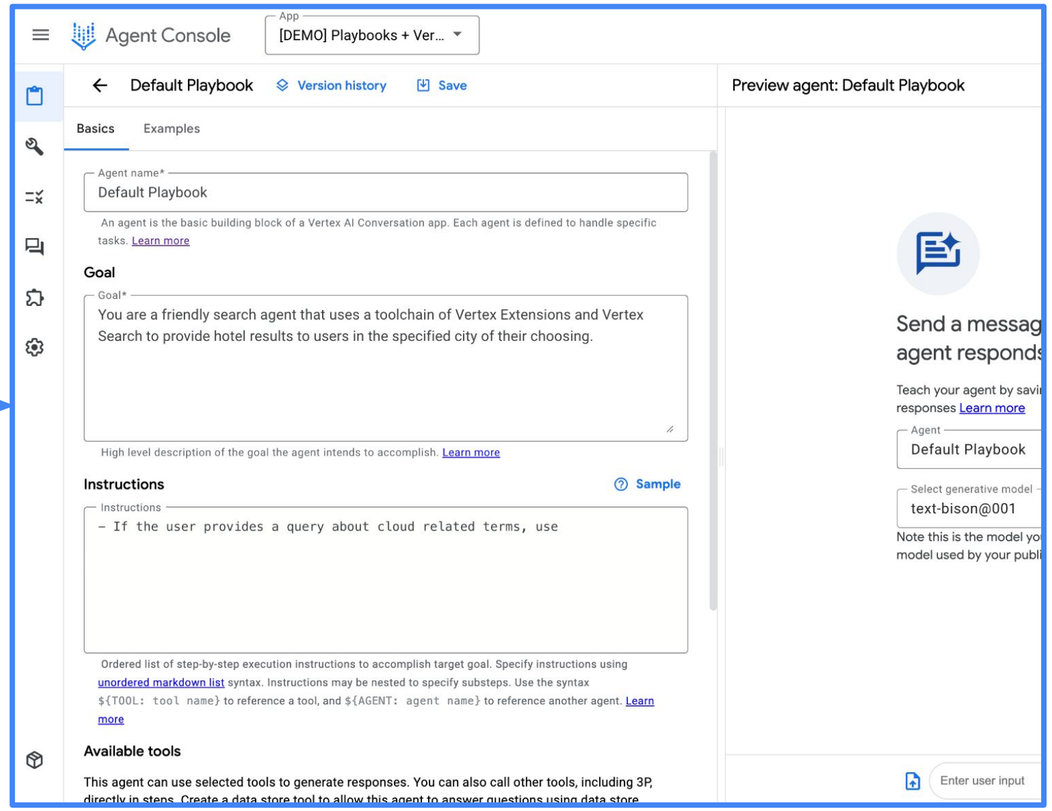
Today's surfaces

Conversational Agents Console





AI Applications

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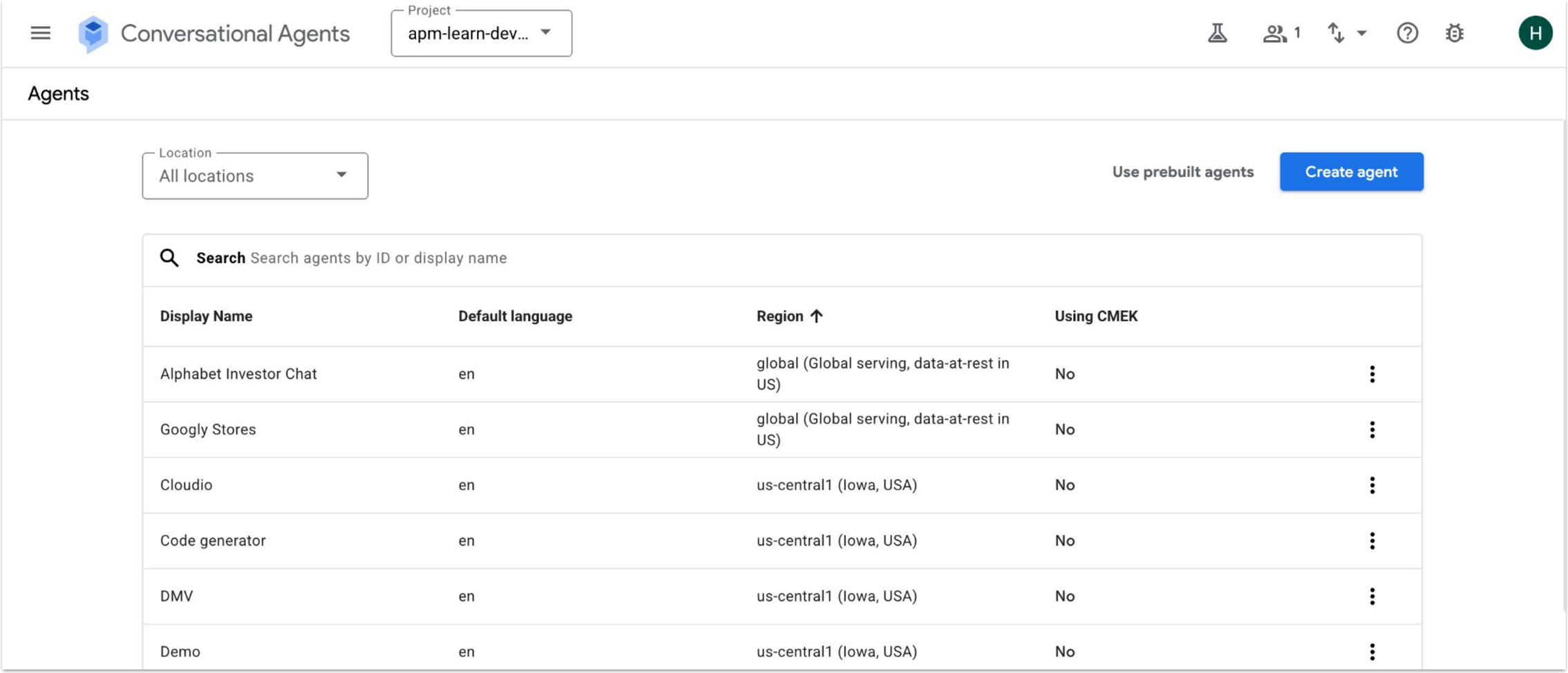
Conversational Agents Console



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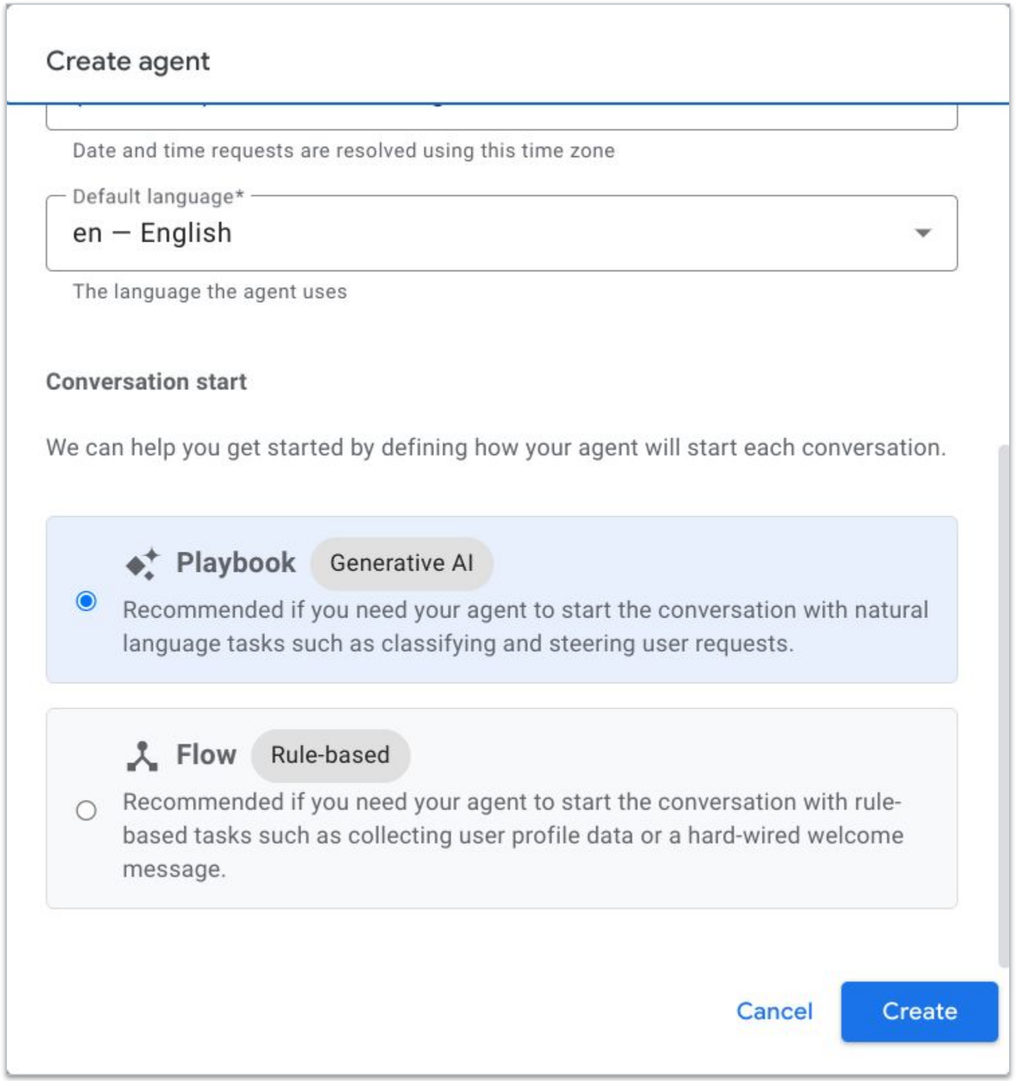
Creating Conversational Agents



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Conversational Agents Console

Conversational Agents | Project: df-decibel2-dev-... | Agent: Online Furniture ... | Language: en

BUILD
- Playbooks
- Flows
- Tools
- Prebuilt

TEST & EVALUATE
- Test Cases
- Conversation history

DEPLOY
- Environments
- Versions

INTEGRATION
- Integrations
- Conversation Profiles

online furniture bot | Version history | Save

Basics | Examples | Settings

Playbook name*: online furniture bot

A playbook is the basic building block of a Vertex AI Conversation app. Each playbook is defined to handle specific tasks. Learn more

Goal

Goal*: A friendly online furniture store assistant

High level description of the goal the playbook intends to accomplish. Learn more

Instructions | Sample

Instructions:
```
- use ${FLOW:order handling} to handle a new or existing order
- use ${FLOW:arrange appointment} to schedule an appointment
- use ${TOOL:Ikea website} to answer user questions
```

Ordered list of step-by-step execution instructions to accomplish target goal. Specify instructions using unordered markdown list syntax. Instructions may be nested to specify substeps. Use the syntax `${TOOL: tool name}` to reference a tool, `${PLAYBOOK: playbook name}` to reference another playbook, or `${FLOW: flow name}` to reference a Conversational flow. Learn more

+ Add code block You can define inline actions using the `@Action` decorator and handlers using decorators like `@EventHandler`. These allow you to control your agent's behavior using python code.

Available tools

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+ Data store | Manage all tools

- [ ] ice cream shop FAQ (Data store)

Edit example | Save | Cancel

Examples are sample conversations between an end-user and the playbook, including the dialogue and the agent's corresponding actions. Learn more

Display name*: place order example2

Basic Settings

Input & Output

- hello
- Hi there! How can I help you today?
- move to an apartment and want to furnish
- Sorry something went wrong, can you repeat?
- what products do you sell
- Ikea website (Tool) | Ikea website (Action) | 1 Input parameters | 1 Output parameters
- Furniture: https://www.ikea.com/us/en/cat/furniture-10066/
  - Home goods: https://www.ikea.com/us/en/cat/home-goods-10067/
  - Accessories: https://www.ikea.com/us/en/cat/accessories-10068/
- I want to place an order for a table
- order handling (Flow) | 0 Input parameters | 0 Output parameters

Enter user input

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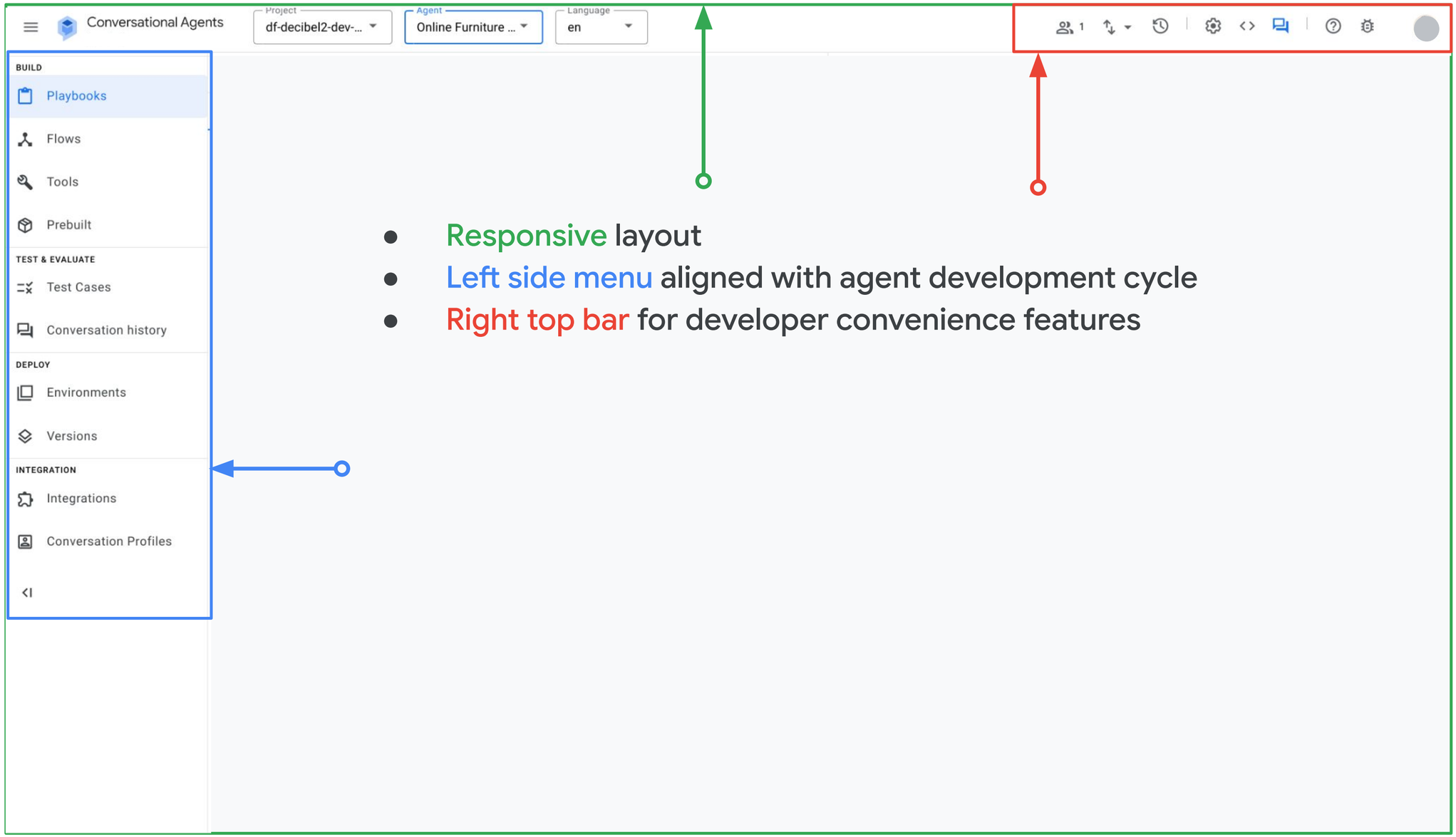
Conversational Agents Console



- Responsive layout
- Left side menu aligned with agent development cycle
- Right top bar for developer convenience features

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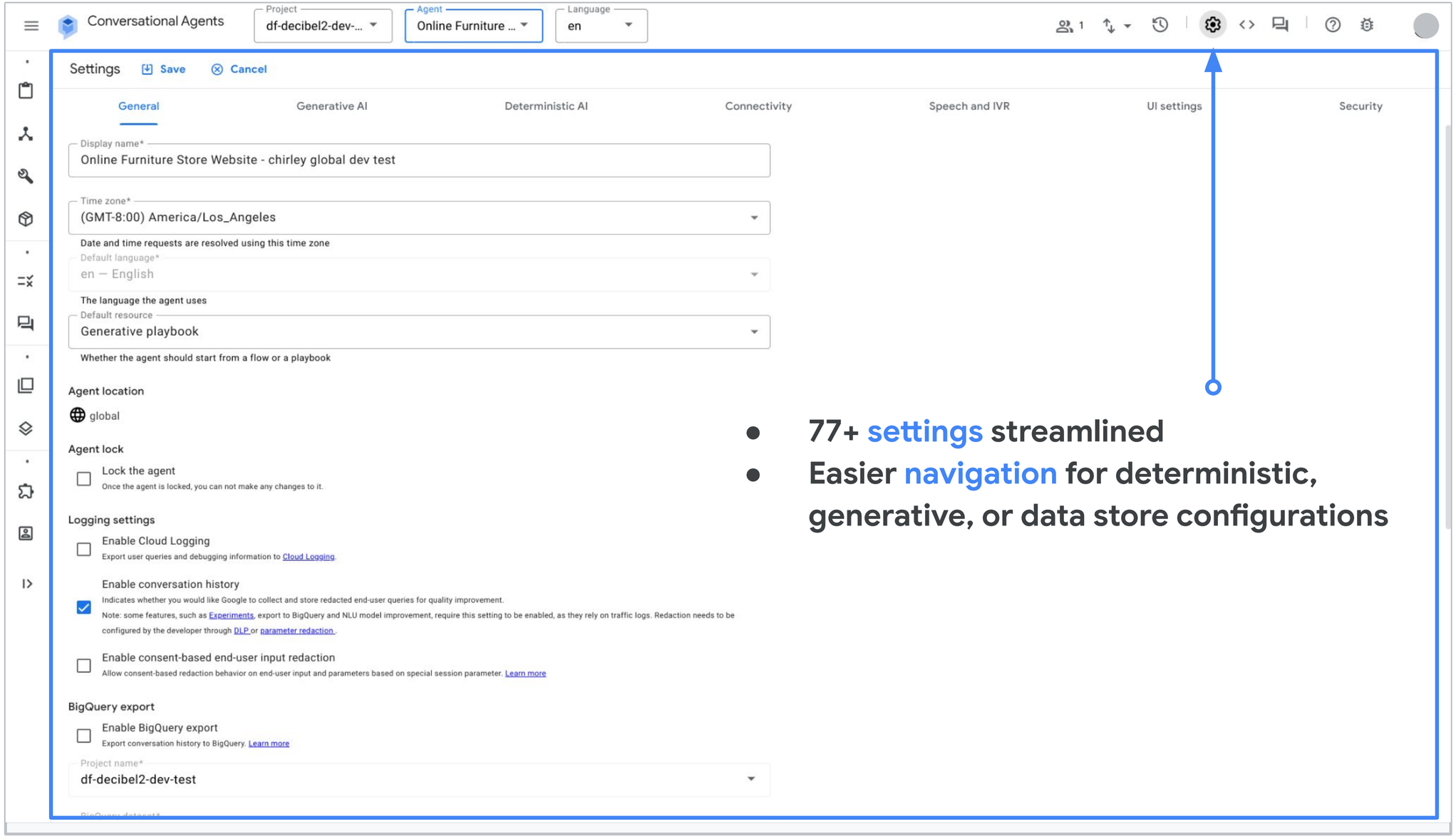
Conversational Agents Console



- 77+ settings streamlined
- Easier navigation for deterministic, generative, or data store configurations

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Enter user input

Google Cloud